

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. A genuine natural-extension past, rather than a future-determining stable quotient, carries the prepared history process. We prove a weighted Feller kernel and a quenched enhanced invariance principle. Exact memory is obtained from the full compressed resolvent with the causal sign; instantaneous Dirac terms are separated from an exponentially decaying regular residual, and finite principal parts are realized without discarding the unresolved space.

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1. ROUND-NINE RECONSTRUCTION: QUENCHED PAST KERNELS AND DISTRIBUTIONALLY SEPARATED MEMORY

The first part constructs a Markov/Feller process from the genuine symbolic past and proves a quenched enhanced invariance principle. The second part is an operator theorem for compressed resolvents. Instantaneous Dirac terms are separated before any exponential-decay statement, and the full unresolved space is retained in the realization.

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1.1. The natural-extension past process. Let $(\widehat{\Omega}, \widehat{\sigma}, \widehat{\nu}_\Psi)$ be one recurrent Gibbs component from A3, with countable alphabet, summable-variation potential, and roof τ . A history is

$$h = (\dots, \omega_{-2}, \omega_{-1}, \omega_0, s), \quad 0 \leq s < \tau(\omega_0).$$

Equip histories with

$$d_\beta(h, h') = |s - s'| \wedge 1 + \sum_{j \geq 0} e^{-\beta j} 1_{\{\omega_{-j} \neq \omega'_{-j}\}}$$

and weight $W(h) = 1 + \tau(\omega_0) + \sum_{j \geq 0} e^{-\beta j} r(\omega_{-j})$. Let C_0^W be the weighted uniformly continuous functions vanishing at W -infinity.

The normalized g -function of the natural extension gives the conditional law of the next symbol given the past. Flowing the age coordinate until the roof is reached and then using this kernel defines $P_t(h, \cdot)$.

Theorem 1.1 (Weighted past-kernel Feller semigroup). *The kernels $(P_t)_{t \geq 0}$ satisfy Chapman–Kolmogorov and define a strongly continuous positive contraction semigroup on C_0^W . On every compact history set,*

$$|P_t F(h) - P_t F(h')| \leq C_T \omega_F(C_T d_\beta(h, h')) + C_T \|F\|_W e^{-cL}$$

when the two pasts agree through depth L . The associated resolvent is normal and separating, hence its Ray compactification carries a right-continuous strong Markov realization.

Proof. Summable variations make the one-step g -kernel Lipschitz in the weighted past metric. Couple equal finite pasts with the same future symbols until the coupling fails; the failure probability is bounded by the variation tail. Exponential return tails control the number of renewals before time T and the W -tail uniformly on compact sets. This gives the displayed modulus, Feller continuity, and preservation of C_0^W . Right continuity of the age flow and the renewal tail give strong continuity. Splitting at the first renewal proves Chapman–Kolmogorov. The Laplace resolvent is conservative, normal by monotone convergence, and separates histories through cylinder functions, so the Ray theorem applies without using a process constructed in advance. $\square$

1.2. The oriented conditional Poisson equation. Let A be a centered vector observable in the A2/A3 weighted Hölder class. Write $\mathbf{P}$ for the one-step past-state Markov operator. The spectral gap on the recurrent component gives

$$\chi = \sum_{n \geq 1} \mathbf{P}^n A$$

in the weighted Hölder norm. Define

$$D_{n+1} = A(H_n) + \chi(H_{n+1}) - \mathbf{P}\chi(H_n).$$

Lemma 1.2 (Conditional martingale–coboundary decomposition). *For the past filtration $\mathcal{F}_n^- = \sigma(H_0, \dots, H_n)$,*

$$E(D_{n+1} \mid \mathcal{F}_n^-) = 0,$$

and

$$\sum_{j < n} A(H_j) = \sum_{j < n} D_{j+1} + \chi(H_0) - \chi(H_n) + R_n,$$

where the roof-centering remainder R_n has exponential moments uniformly for initial histories in a compact set.

Proof. The definition of $\mathbf{P}$ gives $E(\chi(H_{n+1}) \mid \mathcal{F}_n^-) = \mathbf{P}\chi(H_n)$, proving the martingale property. The Poisson identity $(I - \mathbf{P})\chi = \mathbf{P}A$ gives the telescoping decomposition after one index shift. The suspension interpolation and unfinished final roof segment form R_n ; the A3 pointed-excursion tail supplies its uniform exponential moment. $\square$

Theorem 1.3 (Uniform quenched enhanced invariance principle). *For every compact history set, the conditional laws of*

$$\epsilon \int_0^{t/\epsilon^2} A(H_s) ds$$

and their canonical second iterated integrals converge uniformly to a Brownian rough path. The covariance is the conditional Green–Kubo form and the antisymmetric second-level drift is the absolutely convergent area anomaly. The convergence is stable with respect to bounded initial-history variables.

Proof. Lemma 1.2 reduces the first level to a triangular martingale array. The conditional brackets are additive functions of the past state; the Feller coupling and recurrent spectral gap make their time averages converge uniformly on compact histories. Uniform exponential moments give conditional Lindeberg and tightness. The martingale rough invariance theorem then gives the first and second levels. The coboundary contributes the antisymmetric limit

$$\Gamma = \frac{1}{2} \sum_{n \in \mathbb{Z}} (E[A_0 \otimes A_n] - E[A_n \otimes A_0]),$$

which is absolutely convergent. Multiplying by an initial-history test does not alter the conditional estimates, proving stable convergence. $\square$

1.3. Compressed resolvents and instantaneous distributions. Let $\mathcal{H} = L^2(\hat{\nu}_\Psi)$, let L be the closed Doob generator, and let P be an arbitrary finite-rank orthogonal projection with $P\mathcal{H} \subset D(L^{M+2})$. Put $Q = I - P$ and

$$C(z) = P(z - L)^{-1}P, \quad \widehat{K}(z) = zP - PLP - C(z)^{-1}.$$

The exact full unresolved realization is the block operator

$$L = \begin{pmatrix} PLP & PLQ \\ QLP & QLQ \end{pmatrix} \quad \text{on } P\mathcal{H} \oplus Q\mathcal{H}.$$

Lemma 1.4 (Instantaneous/residual decomposition). *The high-frequency expansion of $C(z)$ determines a polynomial $\mathcal{P}_M(z)$ such that*

$$\widehat{K}(z) = \mathcal{P}_M(z) + \widehat{K}_{\text{reg}}(z), \quad \|\widehat{K}_{\text{reg}}(-\gamma + ib)\| \leq C(1 + |b|)^{-1-\delta}$$

whenever the $A2$ resolvent has the corresponding strip and derivative bounds. The inverse Laplace transform of $\mathcal{P}_M$ is a finite sum of δ_0 and its derivatives; only K_{reg} is called the decaying memory function.

Proof. Expand $(z - L)^{-1}$ on $P\mathcal{H}$ through order $M + 2$ and invert the finite matrix series for $C(z)$. Nonnegative powers of z and constants form $\mathcal{P}_M$. The declared graph-domain resolvent estimates allow $M + 2$ integrations by parts on a vertical line, leaving the stated integrable decay. Polynomial Laplace transforms are exactly instantaneous distributions, so they cannot be included in an ordinary exponential-decay claim. $\square$

Theorem 1.5 (Exact Volterra equation and exponential regular memory). *The resolved correlation $C(t) = Pe^{tL}P$ satisfies, in causal distributions,*

$$C'(t) = PLPC(t) + (K * C)(t), \quad C(0) = P,$$

with

$$\widehat{K}(z) = zP - PLP - C(z)^{-1}.$$

After the instantaneous part of Lemma 1.4 is removed,

$$\|K_{\text{reg}}(t)\| \leq Ce^{-\gamma t}.$$

Proof. Laplace transformation of the Volterra equation gives $zC(z) - P = PLPC(z) + \widehat{K}(z)C(z)$, which is equivalent to the displayed Schur formula. For the regular part, shift the Bromwich contour to $\Re z = -\gamma$; the integrable vertical bound from Lemma 1.4 justifies the shift and yields the exponential factor. $\square$

1.4. Finite meromorphic modes without passivity. Suppose the finite matrix function $\widehat{K}(z)$ has finitely many poles in the strip after the instantaneous polynomial is removed. Its principal part is rational.

Theorem 1.6 (Descriptor realization of finite principal parts). *There are a finite-dimensional space Z and a regular matrix pencil $zE_Z - A_Z$ such that*

$$\widehat{K}(z) = \mathcal{P}_M(z) + C_Z(zE_Z - A_Z)^{-1}B_Z + \widehat{K}_{\text{dec}}(z),$$

where $\widehat{K}_{\text{dec}}$ is analytic with integrable vertical decay in the strip. The complete state realization is

$$P\mathcal{H} \oplus Q\mathcal{H} \oplus Z;$$

the Q -space is never replaced by Z . No positivity, passivity, or claim that a transmission zero is an eigenmode of QLQ is used.

Proof. Write each matrix principal part in partial fractions and realize every Jordan term $(z - z_j)^{-k}M_{j,k}$ by a finite companion block in a descriptor pencil. Their direct sum gives Z, E_Z, A_Z, B_Z, C_Z . Subtracting the exact principal part leaves an analytic function with the vertical decay of Theorem 1.5. Adjoining this finite input-output system to the already exact block realization on $P\mathcal{H} \oplus Q\mathcal{H}$ gives the stated descriptor realization. It is an operator identity, not a unitary microscopic decomposition. $\square$

1.5. Fixed-phase nonlinear history pressure. For a bounded history source V , the Feynman–Kac semigroup on C_0^W is

$$P_t^V F(h) = E_h \left[e^{\int_0^t V(H_s) ds} F(H_t) \right].$$

Define $\mathcal{E}_t^V F = \log P_t^V(e^F) - \log P_t^V 1$.

Theorem 1.7 (History pressure and rough tangent). *The family $(\mathcal{E}_t^V)$ is a strongly continuous nonlinear semigroup on bounded subsets of C_0^W . Its cylinder generator is the exponential transform of the past-kernel generator. Under diffusive scaling it converges, conditionally on compact pasts, to the risk-sensitive diffusion semigroup driven by Theorem 1.3, including the area anomaly. The fixed-phase memory is the one in Theorems 1.5–1.6; no generic Doob conjugacy invariance is asserted.*

Proof. The weighted Feller theorem gives the linear Feynman–Kac tower and strong continuity; exponentiation gives the nonlinear tower. Differentiating on the cylinder core yields the exponential generator. Uniform conditional rough convergence and exponential moments pass the Feynman–Kac functionals to the rough diffusion. The memory statement is fixed in one Hilbert phase, so its pairing and resolved projection do not change. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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